

3-Mark Questions & Answers — DBMS Module 6: Foundations of IBM Db2

Q1. Compare IBM Db2 with Oracle and MySQL.

Feature	IBM Db2	Oracle Database	MySQL
High Availability	Db2 pureScale / HADR	Oracle RAC	Replication / InnoDB Cluster
Analytics Engine	In-Memory BLU Acceleration	In-Memory Option	External analytics engines
Enterprise Focus	Hybrid Cloud, HTAP, IBM Z	Enterprise OLTP	Web apps & microservices

Q2. Explain Db2 system requirements and installation steps.

- **System Requirements:** 64-bit OS (Linux, Windows Server, AIX), 2GB+ RAM, 5GB+ Disk space.
- **Installation Steps:** Launch `db2setup` -> Accept License -> Choose Installation Type -> Create Db2 Instance (`db2inst1`) -> Configure Admin User -> Verify via `db2start` .

Q3. Explain IBM Data Studio and Db2 Web Console interfaces.

- **IBM Data Studio:** Eclipse-based GUI IDE for database development, stored procedure creation, and SQL debugging.
- **Db2 Web Console:** Browser-based monitoring dashboard for memory usage, active connections, and query performance tracking.

Q4. Explain RUNSTATS and REORG utilities in Db2.

- **RUNSTATS** : Collects statistics on tables and indexes to update the system catalog for optimal query planning.
- **REORG** : Defragments table data and index pages to restore physical contiguous storage efficiency.

Q5. Explain Db2 on Cloud and Db2 Warehouse.

- **Db2 on Cloud:** Fully managed database-as-a-service (DBaaS) with automated backups and 99.99% SLA.
- **Db2 Warehouse on Cloud:** In-memory columnar data warehouse powered by BLU Acceleration for large-scale analytics.